Q1 - 25 July - Shift 2

The maximum error in the measurement of resistance, current and time for which current flows in an electrical circuit are 1%, 2% and 3% respectively. The maximum percentage error in the detection of the dissipated heat will be:

(A) 2 (B) 4

(C) 6 (D) 8

Space for your notes:

Q2 - 26 July - Shift 2

If $\vec{A}=\left(2\hat{i}+3\hat{j}-\hat{k}\right)$ m and $\vec{B}=\left(\hat{i}+2\hat{j}+2\hat{k}\right)$ m. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be________ m.

Space for your notes:

Q3 - 27 July - Shift 1

The one division of main scale of vernier callipers reads 1 mm and 10 divisions of Vernier scale is equal to the 9 divisions on main scale. When the two jaws of the instrument touch each other the zero of the Vernier lies to the right of zero of the main scale and its fourth division coincides with a main scale division. When a spherical bob is tightly placed between the two jaws, the zero of the Vernier scale lies in between 4.1 cm and 4.2 cm and 6^{th} Vernier division coincides with a main scale division. The diameter of the bob will be_____ 10^{-2} cm

Space for your notes:

Q4 - 28 July - Shift 1

If the projection of $2\hat{i}+4\hat{j}-2\hat{k}$ on $\hat{i}+2\hat{j}+\alpha\hat{k}$ is zero. Then, the value of α will be

Space for your notes:

Q5 - 28 July - Shift 2

In an experiment to find acceleration due to gravity (g) using simple pendulum, time period of 0.5 s is measured from time of 100 oscillation with a watch of 1s resolution. If measured value of length is 10 cm known to 1mm accuracy. The accuracy in the determination of g is found to be x %. The value of x is

Space for your notes:

Answer Key

Q1 (D) **Q2** (2) **Q3** (412) **Q4** (5)

Q5 (5)

Q1 (D)

$$E_H = I^2R \times t$$

$$\frac{\Delta E}{E}\times 100 = \frac{2\Delta I}{I}\times 100 + \frac{\Delta R}{R}\times 100 + \frac{\Delta T}{T}\times 100$$

$$= 2 \times 2 + 1 + 3 = 8$$

Q2 (2)

If $\vec{A} = \left(2\hat{i} + 3\hat{j} - \hat{k}\right)$ m and $\vec{B} = \left(\hat{i} + 2\hat{j} + 2\hat{k}\right)$ m. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be________ m.

Q3 (412)

10 VSD = 9 MSD

1 VST = .9 MSD

L.C. = .1 mm = .01 cm

+ve zero error = .4mm

= 0.04 cm

Negative zero error = 4.1 cm + 6 × .01

= 4.12 cm

$= 412 \times 10^{-2}$ cm

Q4 (5)

$$\vec{a} \cdot \vec{b} = 0$$

$$\therefore \ \vec{a} \cdot \vec{b} = 0$$

$$\therefore \ 2 \times 1 + 4 \times 2 - 2 \times \alpha = 0$$

$$\therefore \ \alpha = \boxed{5}$$

Q5 (5)

To practice more chapter-wise JEE Main PYQs, click here to download the MARKS app from Playstore

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$g = \frac{1}{4\pi^2}\frac{T^2}{\ell}$$

$$\frac{\Delta g}{g} = \frac{2\Delta T}{T} + \frac{\Delta\ell}{\ell}$$

$$\frac{\Delta g}{g} = 2\cdot\frac{1}{100\times 0.5} + \frac{1mm}{10cm}$$

$$\frac{\Delta g}{g} = \frac{5}{100}$$